

IoT aur OT ethical hacking 1. IoT kya hota hai?

IoT = Internet of Things

Aise physical devices jo **network/Internet se connected** hote hain aur data send/receive kar sakte hain.

Simple examples

- 📺 Wi-Fi CCTV camera
- 🔌 Smart plug
- 💡 Smart bulb
- 🚪 Smart door lock
- ⌚ Smart watch
- 📺 Smart TV
- 🏠 Smart AC
- 🚗 Connected car systems

Example:

Mobile App



Wi-Fi / Internet



Smart CCTV Camera

Aap mobile se camera dekh sakte ho kyunki camera network se connected hai.

2. OT kya hota hai?

OT = Operational Technology

OT ka use **physical machines/processes ko monitor aur control** karne ke liye hota hai.

Examples:

- Factory machines
- Power plants
- Water treatment plants
- Oil & gas systems
- Elevators
- Manufacturing lines
- Industrial robots

Simple example:

Temperature Sensor



PLC



Motor / Valve



Physical Process

Yahan computer/network sirf data nahi de raha—**real-world machine ko control kar raha hai.**

3. IoT vs OT — easiest difference

IoT

Internet-connected devices

Smart camera

Smart bulb

Smart lock

Smart thermostat

Usually consumer/business devices

OT

Industrial/physical systems

Factory machine

Industrial motor

Industrial valve

PLC

Usually industrial infrastructure

IoT

Main focus: connectivity/data

OT

Main focus: physical process/control

Ek line mein:

IoT = Internet se connected चीज़ें

OT = Physical machines/process ko control karne wali technology

4. Ethical Hacker ke liye important kyun hai?

Traditional IT hacking mein attacker generally:

Computer → Data

IoT mein:

Device → Network → Data

OT mein:

Network → Controller → Machine → Physical World

Isliye OT security mein ek compromised device ka impact **physical** bhi ho sakta hai.

Example:

Agar office ka smart bulb compromise ho gaya:

💡 Bulb ON/OFF

Lekin industrial system mein unauthorized command:

Controller → Motor/Valve → Physical process change

Isliye OT security mein **safety aur availability** bahut important hoti hai.

5. IoT Ethical Hacking ka practical example

Maan lo tumhare paas apna **Wi-Fi smart camera** hai.

Ethical hacker normally ye questions check karega:

Step 1 — Device discover karna

Apne authorized lab network mein:

```
nmap -sn 192.168.1.0/24
```

Isse network par active devices identify kar sakte ho.

Example:

192.168.1.1 Router

192.168.1.10 Laptop

192.168.1.20 Smart Camera

192.168.1.30 Smart TV

6. Camera ke open ports check karna

Authorized device par:

```
nmap -sV 192.168.1.20
```

Example result:

PORT	STATE	SERVICE
------	-------	---------

80/tcp	open	http
--------	------	------

443/tcp	open	https
---------	------	-------

554/tcp	open	rtsp
---------	------	------

Beginner ke liye iska matlab:

- **80** → Web interface
 - **443** → HTTPS web interface
 - **554** → RTSP video streaming
-

7. Ethical hacker kya investigate karega?

Ab attacker banne ke bajay **security tester** ki tarah socho:

Questions:

1. Default password hai?

Example:

admin / admin

admin / password

Agar default credentials change nahi kiye gaye → security finding.

2. HTTP use ho raha hai?

Agar sensitive login plain HTTP par ho raha hai → investigate.

3. Firmware outdated hai?

Manufacturer ke security advisories/checksum/version information se compare.

4. Unnecessary services enabled hain?

Example:

FTP

Telnet

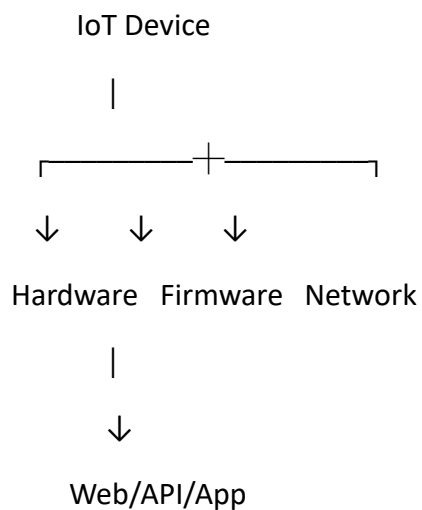
HTTP

RTSP

Agar unnecessary service running hai → attack surface increase.

8. IoT attack surface

Ethical hacking mein tum device ko different layers par examine kar sakte ho:



Hardware

Check:

- Debug interfaces
- USB/UART/JTAG
- Physical access controls

Firmware

Check:

- Hardcoded credentials
- Old libraries
- Secrets
- Vulnerable components

Network

Check:

- Open ports
- Weak protocols
- Unencrypted communication

Web/API

Check:

- Authentication
- Authorization
- Input validation
- API security

9. OT ko practical example se samjho

Suppose ek small factory hai.

SCADA Computer



PLC



Motor Valve Sensor

PLC kya hai?

PLC = Programmable Logic Controller

PLC basically industrial computer/controller hai jo machines ko control karta hai.

Example:

Sensor → PLC → Motor

Sensor bolta hai:

Temperature high hai.

PLC programmed logic ke according:

Motor OFF karo.

10. SCADA kya hota hai?

SCADA = Supervisory Control and Data Acquisition

Simple language mein:

SCADA ek system hai jisse operator industrial process ko monitor/control karta hai.

Example screen:

```
-----  
FACTORY CONTROL  
-----
```

Temperature: 72°C

Motor 1: ON

Pump 1: ON

Valve 1: 40%

Status: NORMAL

Operator isi interface se process monitor karta hai.

11. OT Ethical Hacking practical lab

Real factory par testing **mat karna**.

Instead, lab bana sakte ho:

Your Laptop



Virtual OT Network



PLC Simulator



SCADA Simulator

Is environment mein tum safely practice kar sakte ho.

Useful beginner tools:

- **Nmap** → Network discovery
 - **Wireshark** → Packet analysis
 - **Python** → Automation/testing
 - **Modbus simulator** → Industrial protocol practice
 - **OpenPLC** → PLC learning/testing
 - **CODESYS** → PLC/automation concepts
-

12. Modbus ko simple example se samjho

OT mein **Modbus** ek commonly encountered industrial communication protocol hai.

Imagine:

SCADA

|

| Modbus

↓

PLC

|

|— Motor

|— Pump

|— Sensor

SCADA PLC se information request kar sakta hai.

For example:

PLC Register 100

↓

Temperature = 72

Ethical hacker ka objective ho sakta hai:

"Kya unauthorized system PLC se data read kar sakta hai?"

Aur authorized lab mein:

"Kya unauthorized command accept ho rahi hai?"

13. IoT + OT ek saath

Modern factories mein dono combine ho sakte hain.

Example:

IoT Sensors

↓

Network



Gateway



OT Network



SCADA



PLC



Machine

Yahan security boundary bahut important hai.

Agar IoT side compromise hokar OT network tak pahunch jaye, risk significantly increase ho sakta hai.

14. Ethical hacker ka actual workflow

Beginner ke liye ye methodology yaad rakho:

1. Identify



2. Discover



3. Enumerate



4. Analyze



5. Test



6. Document



7. Fix



8. Retest

Example — IoT camera

Identify

Camera IP = 192.168.1.20

Discover

`nmap -sV 192.168.1.20`

Analyze

80 → HTTP

443 → HTTPS

554 → RTSP

Test

Check:

- Authentication
- Encryption
- Firmware
- Access control
- Configuration

Document

Example finding:

Finding:

Default credentials enabled

Risk:

Unauthorized access

Recommendation:

Force password change during setup

15. IoT vs OT hacking mindset

IoT

Focus:

Device



Firmware



Network



Web/API



Cloud

OT

Focus:

Network



Industrial Protocol



PLC

↓

SCADA

↓

Physical Process

OT में खास बात:

"Can I hack it?" से ज्यादा important है "Can my testing disturb the physical process?"

इसीलिए real OT penetration testing बहुत carefully planned होती है।

16. Beginner ke liye practice roadmap

अगर तुम **Ethical Hacking** सीख रहे हो, तो सीधे PLC hacking पर मत जाओ।

Level 1 — Networking

Learn:

IP

MAC

TCP

UDP

Ports

DNS

DHCP

HTTP/HTTPS

Subnetting

Practice:

ping

ipconfig

arp

netstat

nmap

Level 2 — Linux

Learn:

Linux commands

Files

Permissions

Processes

Networking

SSH

Bash

Level 3 — Web Security

Learn:

HTTP

Cookies

Sessions

Authentication

Authorization

APIs

OWASP Top 10

Level 4 — IoT

Learn:

IoT architecture

Firmware

UART/JTAG concepts

MQTT

CoAP

RTSP

IoT APIs

Device security

Level 5 — OT

Learn:

PLC

SCADA

HMI

ICS

Modbus

DNP3

OPC UA

Industrial Ethernet

OT network architecture

Level 6 — OT Security

Then learn:

Network segmentation

Firewalls

ICS monitoring

Asset discovery

Protocol analysis

Vulnerability management

Incident response

Safety considerations

सबसे important difference याद रखो

Imagine **smart bulb**:

 Mobile → Wi-Fi → Bulb

That's **IoT**.

Imagine **factory motor**:

 SCADA → PLC → Motor

That's **OT**.

And imagine:

 IoT Sensor → Network → Gateway → PLC → Motor

That's where **IoT + OT security** becomes especially interesting.

Ethical hacking mein hamesha apne lab, CTF, simulator, ya explicitly authorized system par hi testing karo.